

Model subclassing with keras

Notebook Input Output Logs Comments (1)

```
warnings.filterwarnings('ignore')
```

In [3]:

```
(x_train, y_train), (x_test, y_test) = mnist.load_data()
x_train = x_train.reshape(-1, 28, 28, 1).astype("float32") / 255.0
x_test = x_test.reshape(-1, 28, 28, 1).astype("float32") / 255.0
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11493376/11490434 [=====] - 0s 0us/step
```

In [4]:

```
class CNNBlock(layers.Layer):
    def __init__(self, out_channels, kernel_size=3):
        super(CNNBlock, self).__init__()
        self.conv = layers.Conv2D(out_channels, kernel_size, padding="same")
        self.bn = layers.BatchNormalization()

    def call(self, input_tensor, training=False):
        x = self.conv(input_tensor)
        x = self.bn(x, training=training)
        x = tf.nn.relu(x)
        return x
```

```
model = keras.Sequential(
    [CNNBlock(32), CNNBlock(64), CNNBlock(128), layers.Flatten(), layers.Dense(10),
])
```

```
class ResBlock(layers.Layer):
    def __init__(self, channels):
        super(ResBlock, self).__init__()
        self.channels = channels
        self.cnn1 = CNNBlock(channels[0], 3)
        self.cnn2 = CNNBlock(channels[1], 3)
        self.cnn3 = CNNBlock(channels[2], 3)
        self.pooling = layers.MaxPooling2D()
        self.identity_mapping = layers.Conv2D(channels[1], 3, padding="same")
```

```
call(self, input_tensor, training=False):
x = self.cnn1(input_tensor, training=training)
x = self.cnn2(x, training=training)
x = self.cnn3(x + self.identity_mapping(input_tensor), training=training,
x = self.pooling(x)
return x
```

```
class ResNet_Like(keras.Model):
    def __init__(self, num_classes=10):
        super(ResNet_Like, self).__init__()
        self.block1 = ResBlock([32, 32, 64])
        self.block2 = ResBlock([128, 128, 256])
        self.block3 = ResBlock([128, 256, 512])
        self.pool = layers.GlobalAveragePooling2D()
        self.classifier = layers.Dense(num_classes)
```

```
call(self, input_tensor, training=False):
x = self.block1(input_tensor, training=training)
x = self.block2(x, training=training)
x = self.block3(x, training=training)
x = self.pool(x, training=training)
x = self.classifier(x)
return x
```

```
model(self):  
x = keras.Input(shape=(28, 28, 1))  
return keras.Model(inputs=[x], outputs=self.call(x))
```

```

2021-10-09 12:52:41.8331904: I tensorflow/compiler/jit/xla_gpu_device.cc:41] Not creating XLA devices, tf_xla_enable_xla_devices not
set
2021-10-09 12:52:41.8336081: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcudart.so.11
2021-10-09 12:52:41.881377: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.882480: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1720] Found device 0 with properties:
pciBusID: 0000:00:04.0 name: Tesla P100-PCIE-16GB computeCapability: 6.0
coreClock: 1.3285GHz coreCount: 56 deviceMemorySize: 15.9061B deviceMemoryBandwidth: 681.8861B/s
2021-10-09 12:52:41.882666: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcudart.so.11.0
2021-10-09 12:52:41.885346: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcublas.so.11
2021-10-09 12:52:41.885424: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcublasLt.so.11
2021-10-09 12:52:41.128673: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcufft.so.10
2021-10-09 12:52:41.129459: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcusolver.so.10
2021-10-09 12:52:41.155628: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcusparsesolver.so.10
2021-10-09 12:52:41.163844: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcusparsesparse.so.11
2021-10-09 12:52:41.166621: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcudnn.so.8
2021-10-09 12:52:41.166782: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.167445: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.169259: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1862] Adding visible gpu devices: 0
2021-10-09 12:52:41.170198: I tensorflow/core/platform/cpu_feature_guard.cc:142] This TensorFlow binary is optimized with oneAPI
Deep Neural Network Library (oneDNN) to use the following CPU instructions in performance-critical operations: AVX2 AVX512F
MA
To enable them in other operations, rebuild TensorFlow with the appropriate compiler flags.
2021-10-09 12:52:41.170427: I tensorflow/compiler/jit/xla_gpu_device.cc:99] Not creating XLA devices, tf_xla_enable_xla_devices not
set
2021-10-09 12:52:41.175993: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.171434: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1720] Found device 0 with properties:
pciBusID: 0000:00:04.0 name: Tesla P100-PCIE-16GB computeCapability: 6.0
coreClock: 1.3285GHz coreCount: 56 deviceMemorySize: 15.9061B deviceMemoryBandwidth: 681.8861B/s
2021-10-09 12:52:41.171476: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcudart.so.11.0
2021-10-09 12:52:41.171499: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcublas.so.11
2021-10-09 12:52:41.171516: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcublasLt.so.11
2021-10-09 12:52:41.171532: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcufft.so.10
2021-10-09 12:52:41.171549: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcusolver.so.10
2021-10-09 12:52:41.171564: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcusparsesolver.so.10
2021-10-09 12:52:41.171581: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcusparsesparse.so.11
2021-10-09 12:52:41.171598: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcudnn.so.8
2021-10-09 12:52:41.171679: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.172327: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.172865: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1862] Adding visible gpu devices: 0
2021-10-09 12:52:41.173626: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcudart.so.11.0
2021-10-09 12:52:41.175984: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1261] Device Interconnect StreamExecutor with stre
ngth 1 edge matrix:
2021-10-09 12:52:41.585951: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1267] 0
2021-10-09 12:52:41.585961: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1280] 0: N
2021-10-09 12:52:41.587744: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.588541: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.589176: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:941] successful NUMA node read from Sysfs had
negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
2021-10-09 12:52:41.589726: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1486] Created TensorFlow device (/job:localhost/rep
lica-0/task-0/device:GPU:0 with 14957 MB memory) -> physical GPU (device: 0, name: Tesla P100-PCIE-16GB, pci bus id: 0000:00:04.
0, compute capability: 6.0)

```

In [5]:

```

model = ResNet_Like(), model()
base_input = model.layers[0].input
base_output = model.layers[2].output
output = layers.Dense(10)(layers.Flatten()(base_output))
model = keras.Model(base_input, output)

model.compile(
    optimizer = keras.optimizers.Adam(),
    loss = keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=['accuracy'],
)

model.fit(x_train, y_train, batch_size=64, epochs=50, verbose=2)
model.evaluate(x_test, y_test, batch_size=64, verbose=2)
model.save("pretrained")

```

```
2021-10-09 12:52:43.408661: I tensorflow/compiler/mlir/mlir_graph_optimization_pass.cc:116] None of the MLIR optimization passes
are enabled (registered 2)
2021-10-09 12:52:43.417161: I tensorflow/core/platform/profile_utils/cpu_utils.cc:112] CPU Frequency: 2000150000 Hz
```

Epoch 1/58

```
2021-10-09 12:52:44.459192: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcublas.so.11
2021-10-09 12:52:45.271016: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcublasLt.so.11
2021-10-09 12:52:45.290415: I tensorflow/stream_executor/platform/default/dso_loader.cc:49] Successfully opened dynamic library libcudnn.so.8
```